Designing Cyber-Physical Systems with aDSL: a Domain-Specific Language and Tool Support

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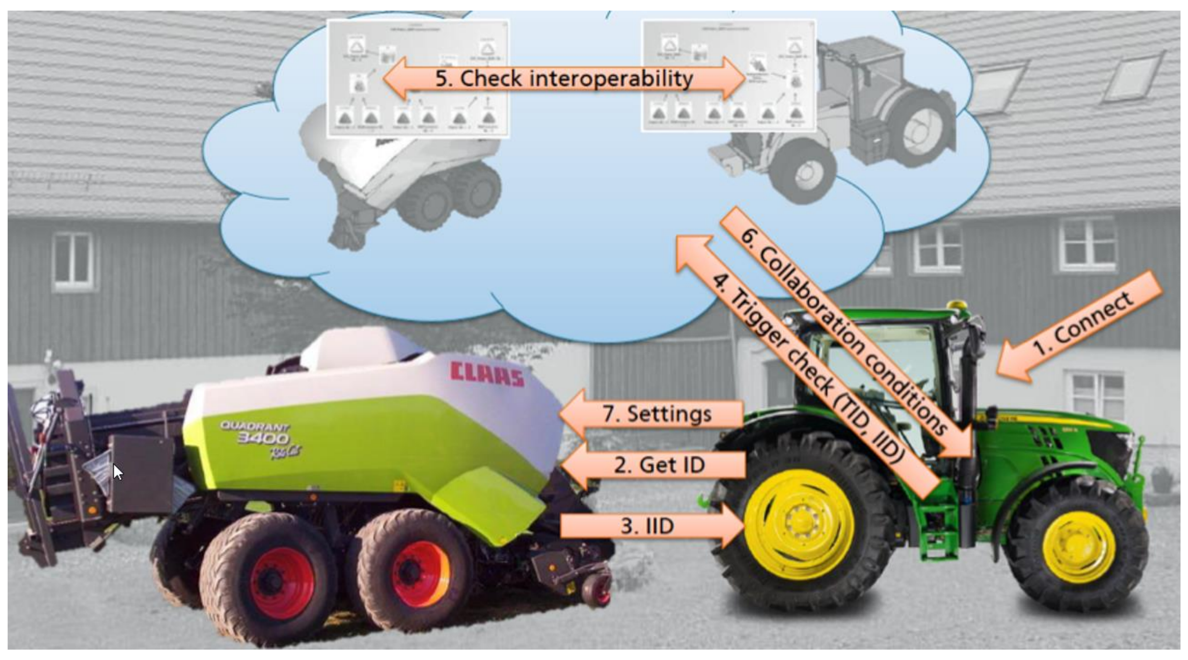
Context & Problem statement

- Cyber-physical systems (CPSs)
  - form an integration of computation, software, networking and physical processes[1]
  - CPS models extend embedded system models with networking, time synchronization, and **interoperability**[2].
  - we propose aDSL, a DSL with tool support for the interoperability of CPSs[3]



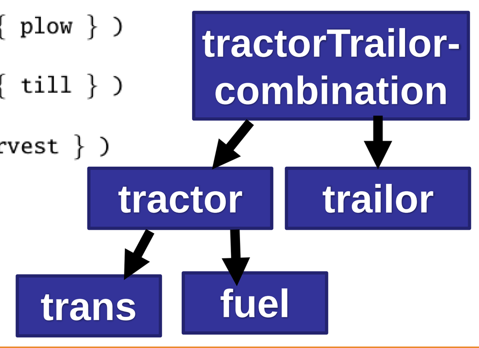
- aDSL objectives:
  - provide a formal language that enables unambiguous definition and reasoning
  - allow for the definition of requirements that are constraints on the operation modes
  - be able to **automatically** evaluate design alternatives

An aDSL instance

The system designer can use the Eclipse IDE for modeling[4]
A CPS comprises systems and parts, each with an operation space.
A requirement constraints these operation spaces

An aDSL system

```
Section system
 Top-level System tractorTrailorCombination OperationSpace () {
   System tractor OperationSpace () {
     System trans OperationSpace () {
      DesAlt (transmission){
        unsynchronized Part transUnsynchronized OperationSpace
            ( driverSkills {advanced moderate easy} continousOperation {no} gears [1 24] )
        doubleClutch Part transDoubleClutch OperationSpace
            ( driverSkills { moderate easy } gears[ 1 2 4])
        CVT Part transCVT OperationSpace
            ( driverSkills {easy} efficiency { frictionLoss } gears [ 1 1000 ] )
      }
     }
     System fuel OperationSpace () {
       DesAlt (engineFuel){
         steam Part fuelSteam OperationSpace
             (pollution [5 10] speed [0 30] )
         diesel Part fuelDiesel OperationSpace
             (pollution [4 8] fuelConsumption [3 5] speed [0 40] price { medium high } )
         gasoline Part fuelGasoline OperationSpace
             (pollution [2 5] fuelConsumption [4 10] speed [0 50] price { medium high })
         electric Part fuelElectric OperationSpace
             (pollution [0 4] fuelConsumption [8 12] speed [0 55] price { high } )
       }
     }
     DesAlt(trailor) {
       chiselPlow Part TrailorChiselPlow OperationSpace
         ( speed [0 25] agility { low veryLow } activity { plow } )
       trailorTiller Part TrailorTiller OperationSpace
         ( speed [0 25] agility { low veryLow } activity { till } )
       chaserBin Part ChaserBin OperationSpace
         ( speed [0 15] agility { veryLow } activity { harvest } )
       notrailor Part NoTrailor OperationSpace
         ( load [0 0] activity { none } )
     }
   }
```

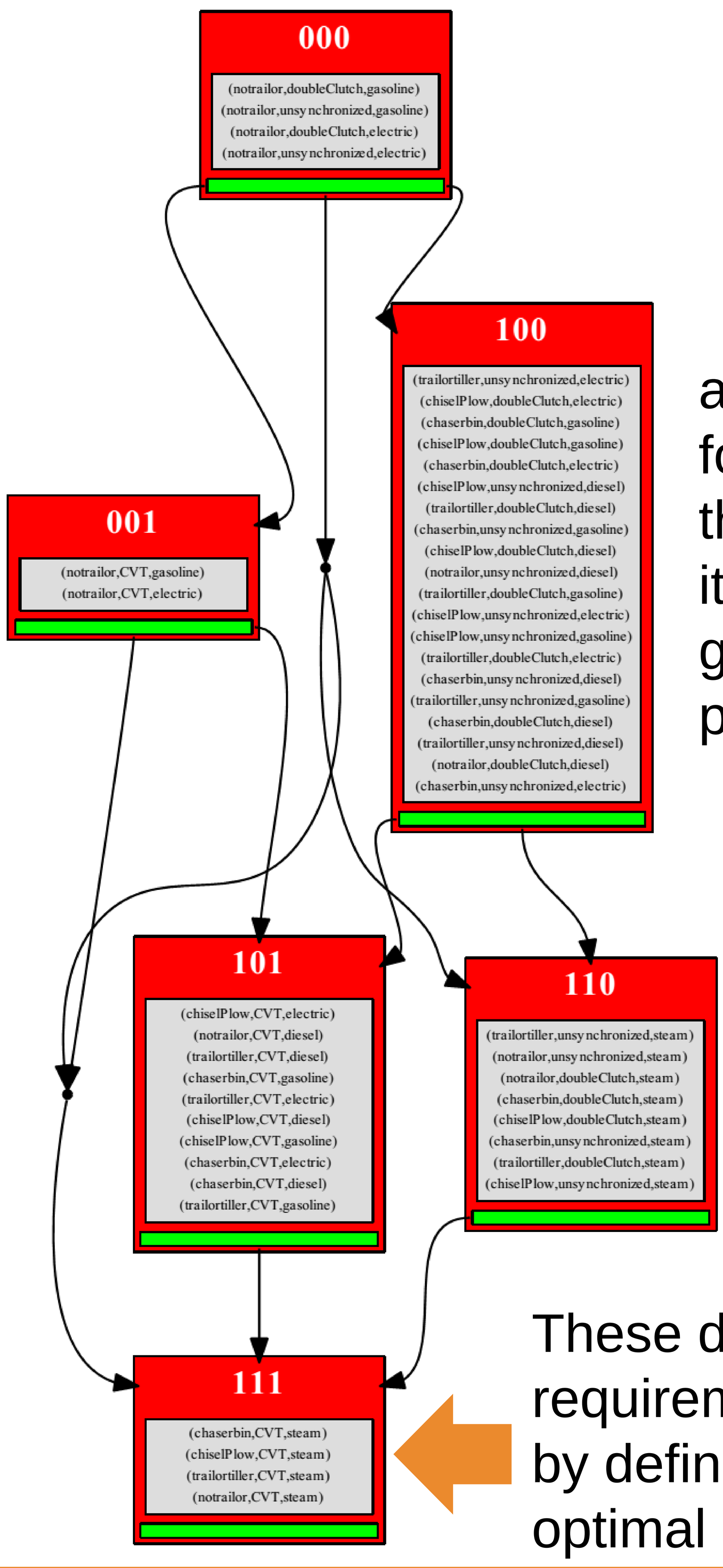


aDSL requirements

```
Section requirements
 Legal Requirement speedRange
   minimum OperationSpace ( speed [5 15] )
   maximum OperationSpace ( speed [0 45] )
 Business Requirement fuelAndPurchaseCosts
   maximum OperationSpace
      ( fuelConsumption [0 20] price {medium})
 Design Requirement operability
   maximum OperationSpace ( driverSkills easy )
```

Pareto optimal designs

We have satisfied the three requirements of aDSL, as mentioned in the objectives



aDSL determines, for each design, the requirements it meets and generates a partial ordering

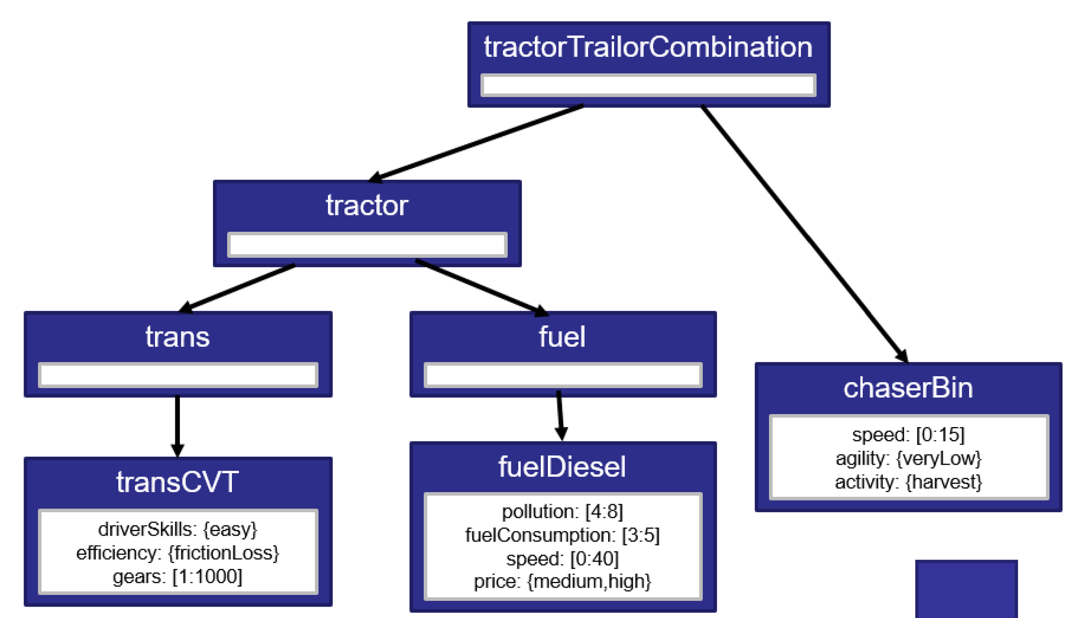
These designs meet requirement 1 and 2 but not 3

These designs meet all requirements and are by definition Pareto optimal

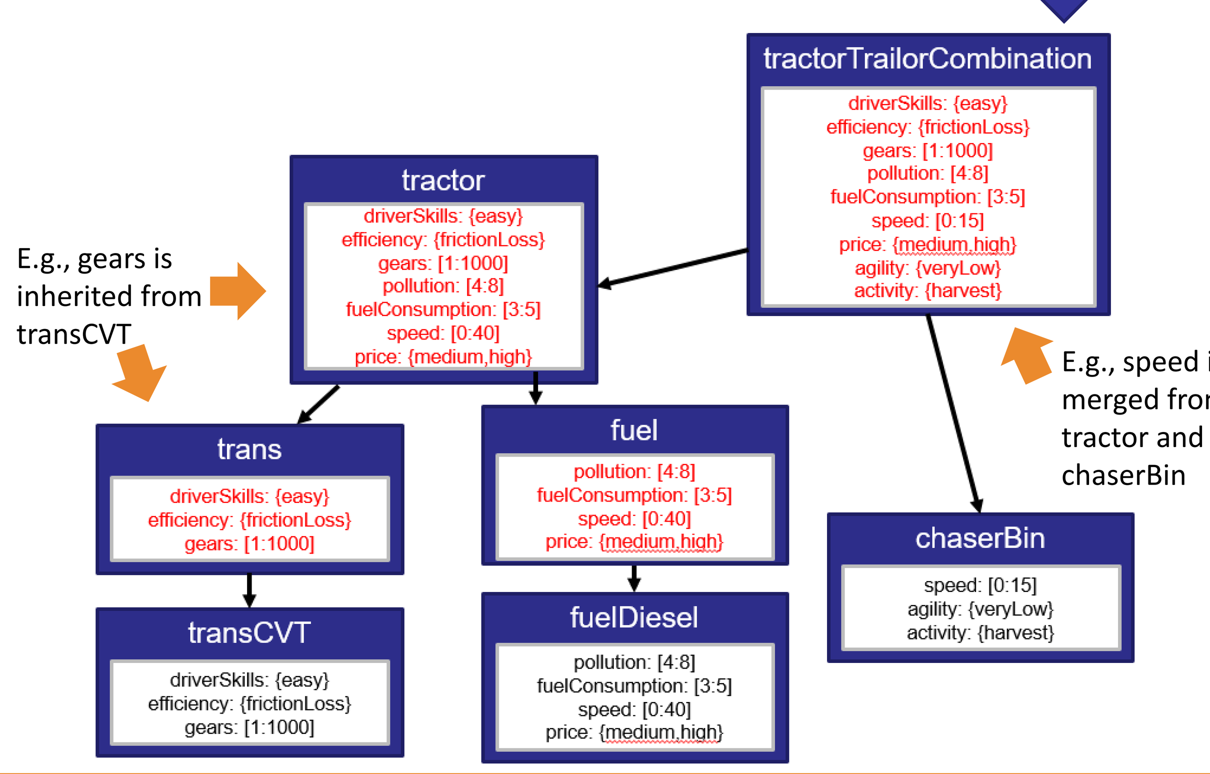
Evaluating operation spaces

aDSL determines the operation space of each subsystem in a **bottom up** fashion; it depends on its operation space and the operation spaces of its children. When operation spaces have overlapping dimensions, the intersection of their values is taken. Otherwise, the dimension and its values are simply copied to a higher level.

Unevaluated aDSL model



Evaluated aDSL model



[1] E. A. Lee, "Cyber-physical systems--Are computing foundations adequate?", Position Paper for NSF Workshop on CPS, 2006.
[2] P. Miller, "Interoperability: What is it and why should I want it?" Ariadne, no. 24, 2000.
[3] F. van den Berg, V. Garousi, B. Tekinerdogan, and B.R. Haverkort, "Designing Cyber-Physical Systems with aDSL: a Domain-Specific Language and Tool Support", 13th System of Systems Engineering Conference, 2018
[4] Eclipse IDE for Java and DSL Developers, http://www.eclipse.org/downloads/packages/eclipse-ide-java-and-dsldevelopers/junosr2

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